

JKBOSE 2026

CLASS 11TH

#XUVV♥

Bhaichara
ON
TOP.

EQUILIBRIUM

LECTURE-4 CHEMISTRY

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Topics To Covered

→ Applications of Keq.

1 Le Chatelier's Principle (Important).

REVISION.

Unit

$$K_c = (\text{mol l}^{-1})^{\Delta n_g}$$

$$K_p = (\text{atm})^{\Delta n_g}$$

K_{eq}

K_p

all substances are gas.

$$K_p = \frac{p_c^c p_d^d}{p_a^a p_b^b}$$



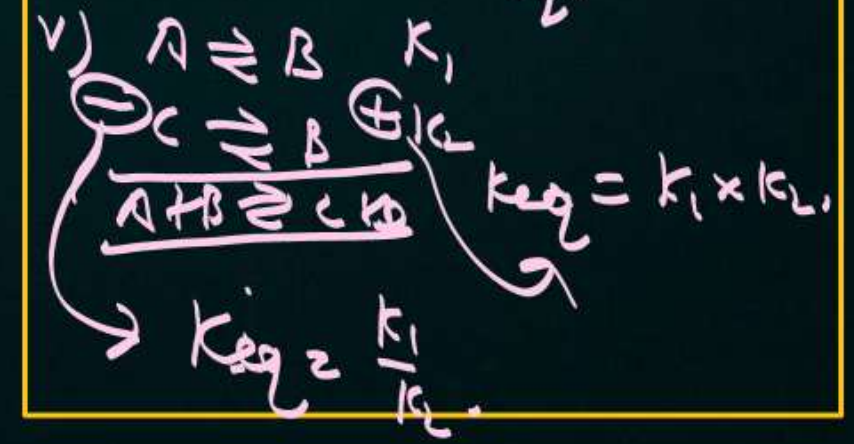
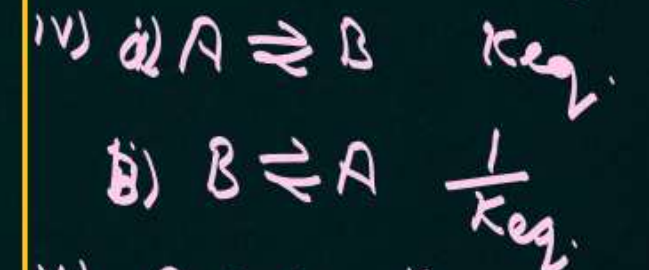
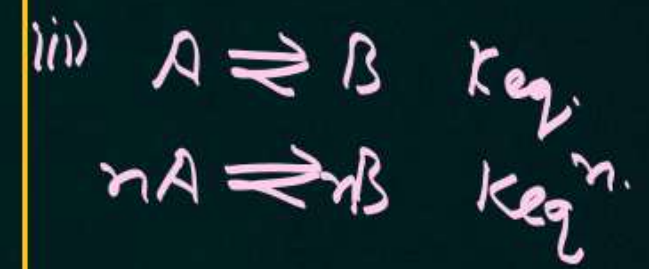
$$K_p = K_c (RT)^{\Delta n_g}$$

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

K_c

K_{eq}

i). only temp dependent



Applications of K_{eq} .

1) TO DETERMINE THE EXTENT OF REACTION $\Rightarrow$

- If the value of K_{eq} is more than 10^3 then it has been found the amount of product is more than Reactant. That reaction is at its final stages.
- If value of K_{eq} is less than 10^{-3} then it has been found that amount of reactant is more than product. That means reaction is at its initial stages.
- If the value of K_{eq} is between 10^3 to 10^{-3} it has been found that both reactant & product have considerable amount. That means reaction is at its middle stages.

2) TO PREDICT THE DIRECTION OF REACTION

To predict the direction of equilibrium we introduce a new term named as Reaction Quotient (Q).

It is also the ratio of molar concentration of products to molar concentration of reactant but at other than equilibrium



$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

It has been found that if

- i) $K_{eq} = Q$, then reaction is in equilibrium
- ii) $K_{eq} > Q$, then reaction will favour forward direction.
- iii) $K_{eq} < Q$, " " " " backward direction.

i) Extent of Reaction

$K_{eq} > 10^3 \rightarrow$ Rxn is about to end.

$K_{eq} < 10^{-3} \rightarrow$ Rxn is at initial stages.

$K_{eq} = 10^3 - 10^{-3} \rightarrow$ middle stages.

ii) Direction of Rxn.

$K_{eq} = Q$, Equilibrium.

$K_{eq} > Q$, forward direction

$K_{eq} < Q$, backward "

EXAMPLE



The value of K_c for the reaction $2A \rightleftharpoons B + C$ is 2×10^{-3} . At a given time, the composition of reaction mixture is $[A] = [B] = [C] = 3 \times 10^{-4} \text{ M}$. In which direction the reaction will proceed?

Sol: Given $K_c = 2 \times 10^{-3}$

$$Q = \frac{[B][C]}{[A]^2} = \frac{3 \times 10^{-4} \times 3 \times 10^{-4}}{(3 \times 10^{-4})^2} = 1$$

$$Q = 1$$

$$\therefore K_c = 2 \times 10^{-3} = 0.002$$

$K_c < Q$, i.e. reaction will proceed in backward direction



$$Q = \frac{[B][C]}{[A]^2} = \frac{[B][C]}{[A][A]}$$

HOT ATTACK

Q. For a reaction that proceeds in forward direction, which option is correct.

a) $K_{eq} = Q$.

b) $K_{eq} > Q$.

c) $K_{eq} < Q$.

d) $K_{eq} = 10^{-3}$.

Le Chatelier's Principle

This principle was given by Louis Le Chatelier a Russian Chemist.

It states that "if a reaction in equilibrium is subjected to change the pressure, concentration & temperature, then equilibrium will shift in that direction which tends to reduce the effect of change".

Applications of Le-Chatelier Principle.

a) Effect of change of concentration

Consider a reaction



If we increase the concentration of any reactant (N_2 or I_2), then equilibrium will shift in that direction where conc. of reactants is decreasing, i.e. forward direction & vice versa. i.e. in short,

Increased concentration of reactants then equilibrium will favour forward direction.

Increased concentration of product, favours backward direction of equilibrium.

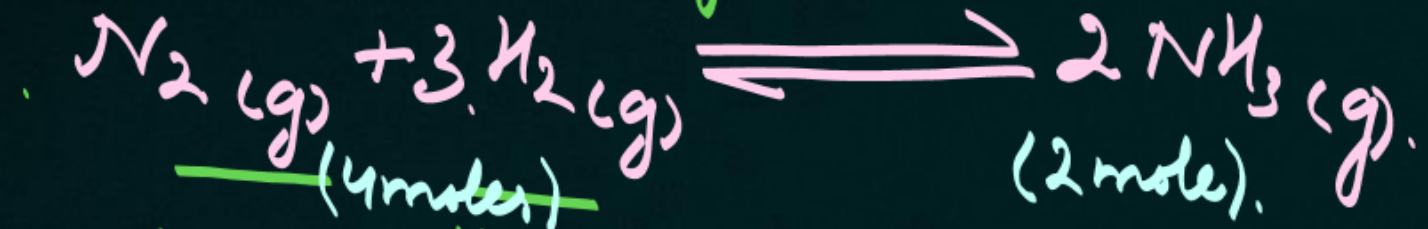


b) Effect of pressure change $\Rightarrow$

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$P \propto \text{no. of moles}$

Consider the following reaction

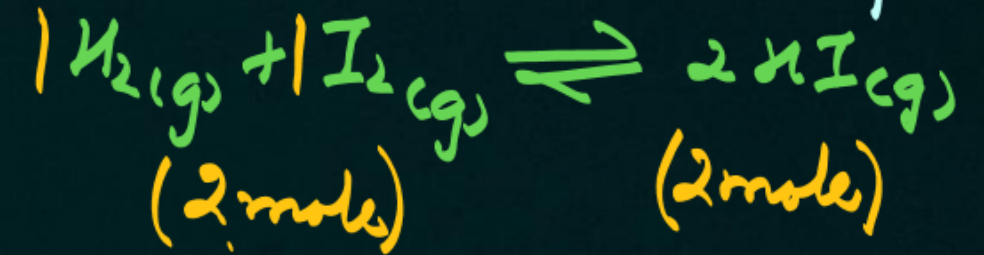


If we increase the pressure of the reaction i.e. we increase the no. of moles, then equilibrium will shift in that direction where less no. of moles are present & vice versa.
In short,

- Increased pressure (no. of moles) will favour the equilibrium direction in which less no. of moles present.
- decreased pressure (no. of moles) will favour the equilibrium direction in which no. of moles are more.

NOTE $\Rightarrow$

If there are equal no. of moles in reactant & products example.



Then, There is no effect of change of pressure on Equilibrium.

Le Chatelier's Principle



C) EFFECT OF TEMPERATURE CHANGE. $\therefore \rightarrow$

Consider the following reaction.



For this reaction, if we increase the temperature by supplying heat, then reaction will shift in that direction where heat is absorbed ($\Delta H = +ve$) i.e. backward direction.

In short.

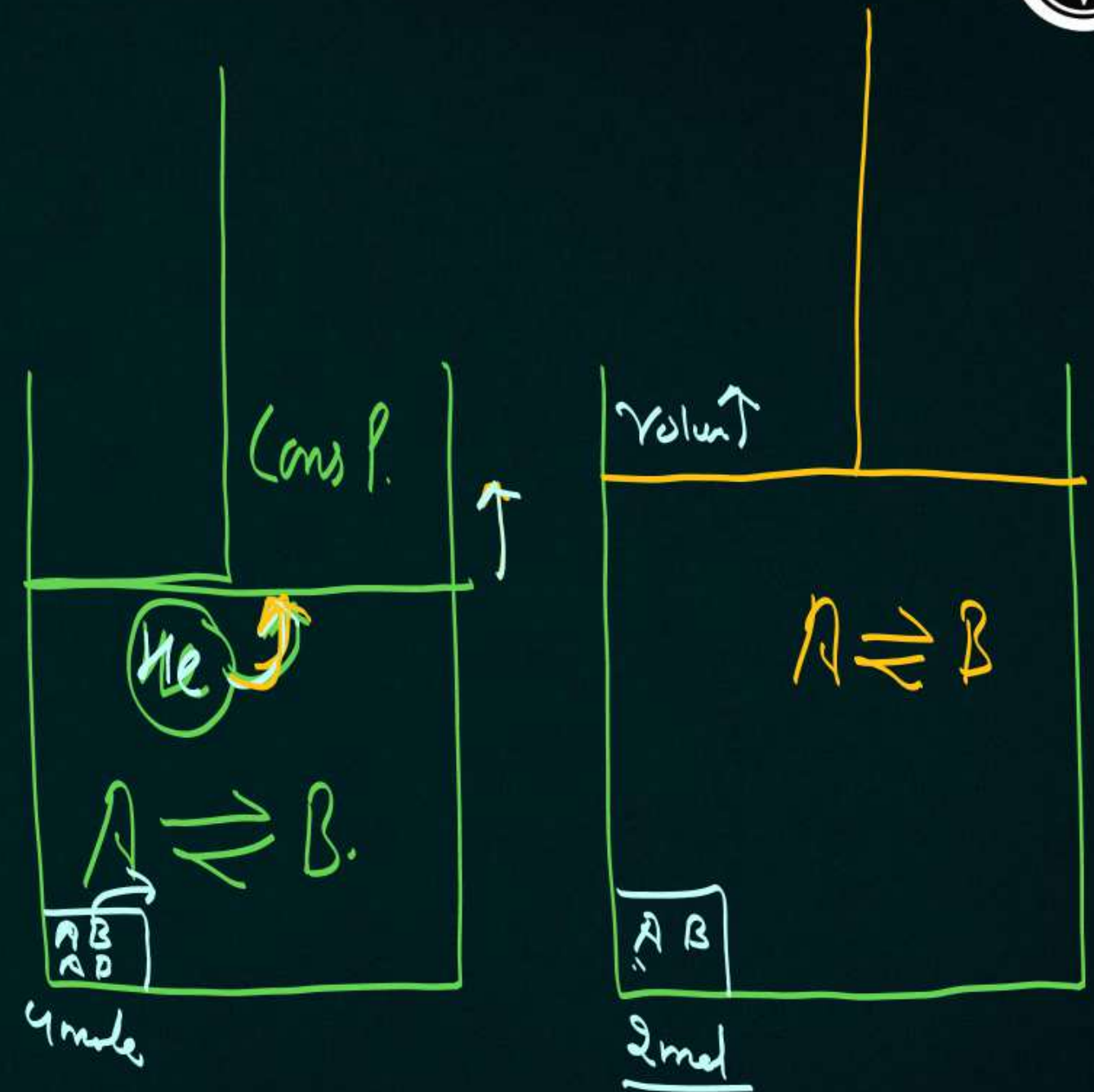
- Increased temperature will favour endothermic reaction.
- Decreased temperature will favour exothermic reaction.

D) EFFECT OF ADDING INERT GAS.

i) At constant volume. If we add inert gas then overall pressure is increased. But this will not change the pressure, concentration or temperature of reaction that means equilibrium will remain unaffected.



iii) At constant pressure $\rightarrow$ On addition of inert gas, increase in volume takes place. If volume increases then no. of moles of gas in one word volume are decreased. We know that if no. of moles of gas are decreased then equilibrium will shift in that direction where no. of moles are more.



QUESTION

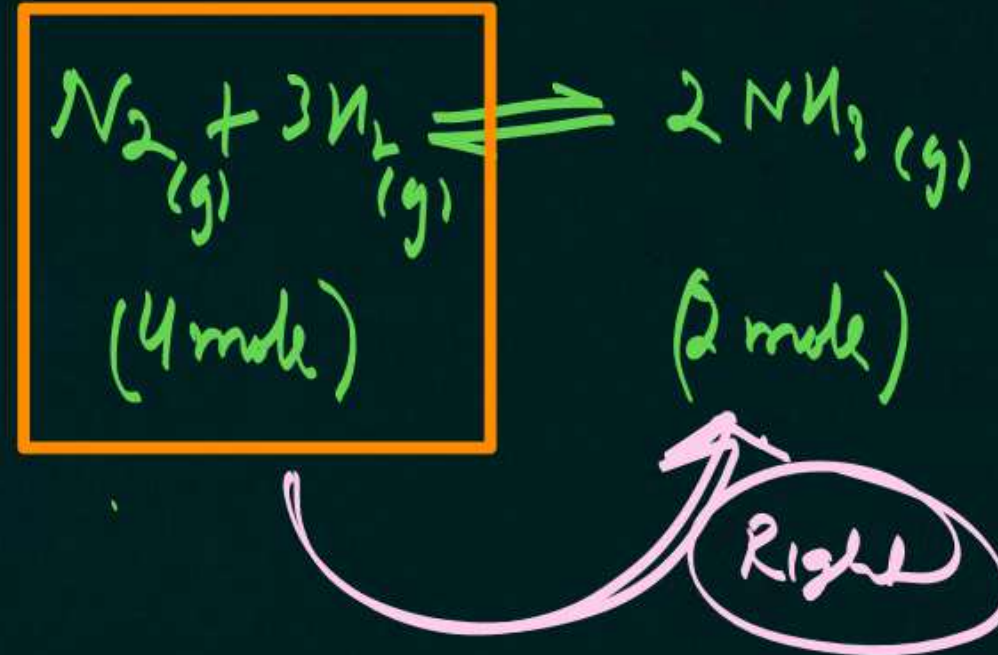
In the reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ Which side equilibrium is shifted when we increase the pressure of the system?

1 Right

2 Left

3 No charge

4 First toward right and then left



QUESTION

In which of the following reactions, the equilibrium remains unaffected on addition of small amount of argon at constant volume?

↳ Inert gas



4 The equilibrium will remain unaffected in all the three cases

QUESTION

Of the following, which change will shift the reaction towards the product at equilibrium?



1 Increase in total pressure (no. of moles) ✗

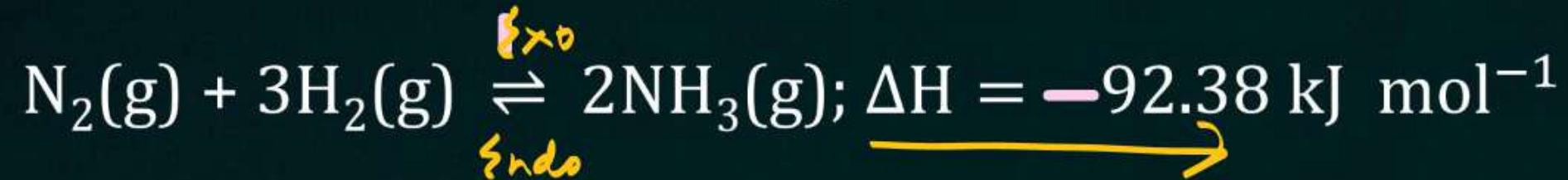
2 Increase in temperature $\uparrow$ Endothermic ✓

3 Decrease in concentration of I_2 ✗ (backward)

4 Increase in concentration of I

QUESTION

Production of ammonia according to the reaction,



$$\Delta H = -ve = \text{exo}$$

$$\Delta H = +ve = \text{endo}$$

is an exothermic process. At low temperature, the reaction shifts in

1

Forward direction

2

Backward direction

3

Either forward or backward direction

4

None of the above

High T $\rightarrow$ Endothermic

Low T $\rightarrow$ Exothermic

- ① Muteeb Riyaz
- ② Mohd. Abbas.
- ③ Dr. Ruband Khan.

Thank
You